Chapter

High-Throughput Glycomics Analysis Enabled by Multiplex Chemical Derivatization and Mass Spectrometry

Miyang Li

Abstract

Glycomics is dedicated to elucidating the structure and function of glycome, which constitutes the full suite of glycans produced by a cell or organism, as well as the regulatory principles behind it. This formidable challenge necessitates the development of robust, sensitive, and scalable technologies for the rapid analysis of numerous glycans, driving the field of high-throughput glycomics. While current methodologies, such as U(H)PLC-FLD, CGE-LIF, and MALDI/LC-MS, have greatly enabled glycomics study, they are often limited by serial sample preparation and sequential analysis, which constrains throughput and introduces inter-batch variability and quantitative inaccuracy. This chapter highlights a critical technological advancement that utilizes the multiplexed isobaric labeling strategy (e.g., aminoxyTMT, SUGAR tags) for analyzing multiple samples by LC-MS/MS at one time. We discuss how this approach dramatically enhances throughput, reduces analytical time and cost, improves data quality by minimizing batch effects, and enables precise relative quantification across sample sets. By focusing on the principles, applications, and future potential of multiplexed glycomics, this chapter underscores its transformative role in advancing glycan profiling for extensive biopharmaceutical studies.

Keywords: glycomics, chemical derivatization, isobaric tags, relative quantification, mass spectrometry

1. Introduction

Glycosylation, the enzymatic attachment of carbohydrates to proteins and lipids, is a ubiquitous and critically important post-translational modification (PTM). It generates a remarkable diversity of glycoconjugates (**Figure 1**), including glycoproteins, proteoglycans, and glycosphingolipids (GSLs), alongside other glycan classes such as free oligosaccharides, glycosaminoglycans (GAGs), and the recently discovered glycosylated RNAs [1]. This structural diversity underpins a vast array of biological functions, influencing protein folding, stability, solubility, immunogenicity, and cellular communication [2]. Given its central role, aberrant glycosylation is

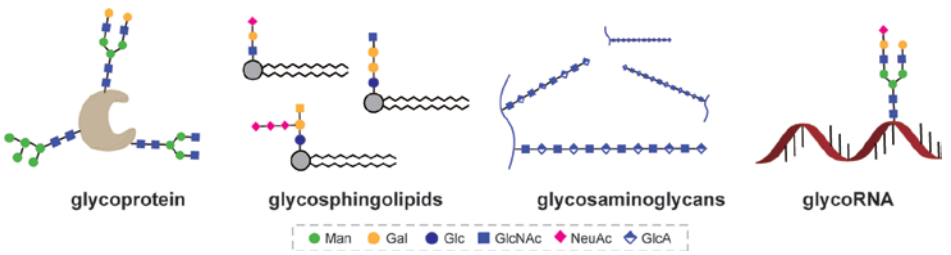


Figure 1.
Schematic representation of major glycan classes, including glycoproteins, glycosphingolipids, proteoglycans/glycosaminoglycans (GAGs), and the recently identified glycoRNAs.

a hallmark of numerous diseases, including cancer, congenital disorders of glycosylation, and autoimmune diseases [3]. Consequently, the comprehensive analysis of the glycome—the complete set of glycans in a biological system—has become a major objective in biomedical research, driving the need for high-throughput analytical methods to decipher glycan functions and discover glycan-based biomarkers in large cohort studies.

Significant progress has been made in high-throughput glycomics, as comprehensively reviewed elsewhere [4]. Core methodologies typically involve the release of glycans (e.g., *via* PNGase F digestion for N-glycans), followed by separation-based techniques like ultra-high-performance liquid chromatography with fluorescence detection (UHPLC-FLD) or capillary gel electrophoresis with laser-induced fluorescence (CGE-LIF), and liquid chromatography-mass spectrometric (LC-MS) analysis. While these platforms have been successfully applied to profile research and clinical samples, they operate predominantly in a serial analysis mode. This presents inherent limitations: analysis time scales linearly with the number of samples, instrument duty cycle is underutilized, and technical variability between processing batches can introduce significant noise, complicating data integration and interpretation in large-scale studies.

A paradigm-shifting solution to these bottlenecks is the adoption of multiplexed isobaric labeling. This approach enables researchers to covalently tag the reducing terminus of glycans from different samples with unique isobaric tags, such as aminoxy tandem mass tags (aminoxyTMT) or isobaric tags for relative and absolute quantitation of glycans (SUGAR). These tags confer identical precursor masses to labeled glycans but yield distinct, quantifiable reporter ions upon collisional dissociation in MS/MS. Consequently, glycans from multiple different samples (currently up to 12-plex) can be pooled and analyzed in a single LC-MS/MS injection. The relative abundance of each glycan across the original samples is then determined by comparing the intensities of the respective reporter ions, dramatically increasing throughput while reducing technical variance and instrument time per sample.

This chapter addresses a critical gap in the literature by focusing on this powerful yet underappreciated strategy. We will elucidate the chemistry behind leading multiplex tags, detail their integration into analysis workflows, and discuss their profound impact on enhancing throughput, quantitative precision, and data quality in high-throughput glycomics. By highlighting the advantages and applications of multiplexed glycan analysis *via* isobaric labeling, we position it as a foundational technique for the next generation of large-scale glycomics studies, poised to accelerate discovery in personalized medicine and biotherapeutic development.

2. Historical overview and methodological evolution in glycomics

Among the diverse array of glycoconjugates, glycoproteins represent a class of exceptional complexity and functional significance. They are primarily classified by their glycosidic linkage: N-linked glycans (attached to the asparagine residue of the canonical Asn-X-Ser/Thr sequon, where X is any amino acid except proline) and O-linked glycans (attached to serine or threonine without a strict consensus sequence). This structural complexity is further defined by two levels of heterogeneity: macroheterogeneity, referring to the occupancy and selection of glycosylation sites on a polypeptide, and microheterogeneity, describing the diverse repertoire of glycan structures that may occupy a single site. Critically, the specific glycoform expressed is not directly genetically encoded but is dynamically regulated by cellular status, including protein conformation, the availability of glycosyltransferases and nucleotide sugar donors, and metabolic flux [5].

Deciphering this heterogeneity is essential for understanding biological function and is particularly crucial in the biopharmaceutical development, where glycosylation directly impacts drug efficacy, stability, and immunogenicity. The analytical strategy employed is dictated by the research question, leading to a hierarchy of methodologies. Intact mass analysis provides the most comprehensive view by characterizing proteoforms with their full complement of PTMs. However, it requires highly purified samples, advanced high-resolution mass spectrometry, and complex data deconvolution, rendering it impractical for high-throughput screens. Middle-down approaches, often used for antibodies (e.g., enzymatic cleavage into Fc/Fab domains), simplify intact analysis to better resolve macroheterogeneity but still struggle with the analytical challenges posed by extensive microheterogeneity [6, 7].

In contrast, bottom-up glycoproteomics involves proteolytic digestion followed by LC-MS/MS to analyze glycopeptides. This powerful approach provides site-specific information on both occupancy (macroheterogeneity) and the attached glycan structures (microheterogeneity). Nevertheless, the sheer structural diversity at each site can create significant analytical complexity, often acting as a bottleneck for throughput. This is frequently mitigated by targeted affinity enrichment strategies for specific glycoproteins or glycopeptides [8].

For large-scale epidemiological and clinical biomarker discovery studies, the analysis of released glycans has emerged as the foremost high-throughput strategy (**Figure 2**). This approach involves the enzymatic or chemical liberation of glycans from their protein scaffolds, enabling the profiling of the global glycome from complex biological matrices without prior protein purification. The release of N-glycans is efficiently achieved with the universal enzyme peptide-N-glycosidase F (PNGase F). The analysis of O-glycans remains more challenging, often relying on chemical methods like reductive β-elimination due to the lack of a broad-specificity enzyme. Since native glycans lack efficient chromophores or fluorophores, they typically require chemical derivatization [9]. Early foundational—though now largely obsolete—methods involved hazardous techniques like radioactive labeling coupled with hydrazinolysis for release, a process hampered by the use of explosive anhydrous hydrazine [10].

A pivotal advancement was the widespread adoption of fluorescent tagging *via* reductive amination. First described by Hase et al. in 1979 for free oligosaccharides [11], this method was coupled with HPLC-fluorescence detection (HPLC-FLD) to achieve vastly improved sensitivity. The reaction involves conjugating the reducing end of the glycan to an amine-containing fluorophore—such as 2-aminobenzamide (2-AB),

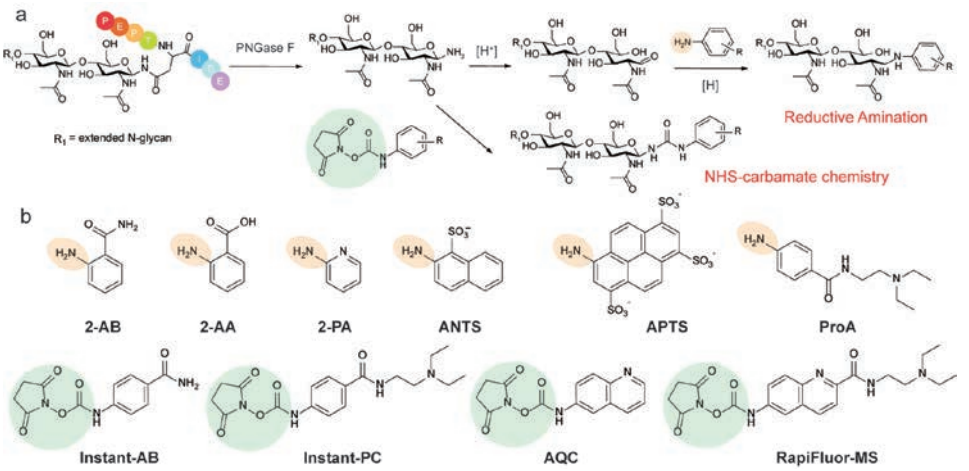


Figure 2.
Evolution of released glycan analysis. (a) Standard workflow for N-glycan release via *PNGase F and subsequent derivatization at the reducing end. (b) Common fluorophores used in glycan reducing-end labeling.*

2-aminopyridine (2-AP), or procainamide (ProA) [12]—using a reducing agent like sodium cyanoborohydride ($NaBH_3CN$) to create a stable derivative. This paradigm enabled sensitive detection from the picomole amounts of material [13] and established the core workflow for high-throughput glycomics.

A more recent innovation aims to bypass the slower reductive amination step by targeting the reactive glycosylamine group immediately after enzymatic release with PNGase F. This approach facilitates rapid labeling with tags such as RapiFluor-MS [14], InstantPC [15], and DiPyrO [16], significantly reducing sample handling time and improving efficiency.

Thus, the historical trajectory of glycomics has been defined by the paradigm of enzymatic release (greatly enabled by PNGase F), chemical derivatization (evolving from hazardous procedures to efficient fluorescent tagging), and chromatographic or electrophoretic separation. This established the robust foundation upon which modern high-throughput glycomics is built, setting the stage for the next revolutionary advance: multiplexed isobaric labeling.

3. Isobaric labeling techniques for high-throughput glycomics

The strategic power of isobaric labeling was first realized in proteomics, where tags like Tandem Mass Tags (TMT) [17] and Isobaric tags for relative and absolute quantitation (iTRAQ) [18] revolutionized quantitative analysis by enabling the simultaneous comparison of multiple samples in a single LC-MS/MS run. This approach minimizes missing values and enhances statistical confidence by reducing run-to-run variability. The fundamental design of these tags involves three key components: [1] a reactive group that targets specific functional groups on analytes (e.g., amines on peptides); [2] a reporter moiety that, upon MS/MS fragmentation, yields a unique low-mass ion for quantification; and [3] a mass balancer that ensures the total mass of the tag is identical (isobaric) across all versions, allowing co-elution and combined MS1 detection [19].

The success in proteomics provided a clear blueprint for adapting this strategy to glycomics. By designing tags that target the reducing end of glycans, researchers have integrated multiplexing into the standard released glycan workflow, creating a powerful paradigm for high-throughput analysis (**Figure 3**). This section details the key isobaric and isotopic tags developed for this purpose (**Figure 4**).

3.1 INLIGHT™

Although not a true isobaric multiplexing technique, INLIGHT (Individuality Normalization when Labeling with Isotopic Glycan Hydrazide Tags) was a foundational advancement in MS-compatible, isotope-coded glycan labeling [20]. This strategy employs a pair of stable isotope-labeled hydrazide tags, enabling duplex quantification based on the area under the curve (AUC) of precursor ions in full MS scans, analogous to the SILAC method in proteomics.

The INLIGHT tags were rationally designed to address the poor ionization efficiency of native glycans. The optimal tag features a hydrophobic biphenyl structure that mitigates ionization bias and enhances ESI-MS sensitivity [21, 22]. A key innovation of the INLIGHT strategy is its sophisticated data processing, which includes

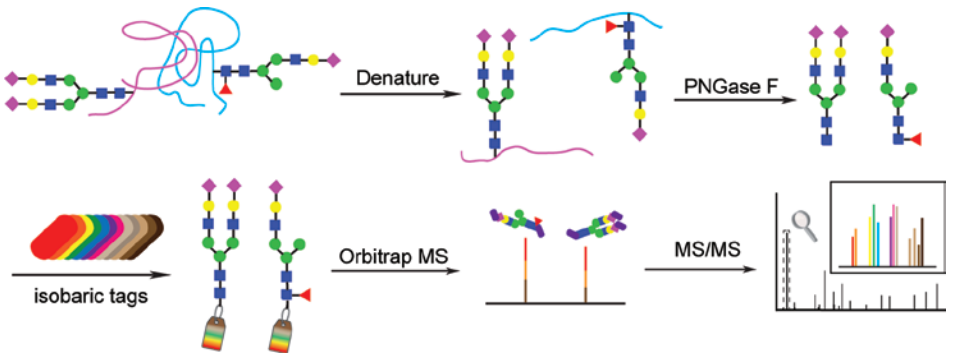


Figure 3.
Workflow for isobaric labeling-enabled high-throughput glycomics. The process involves releasing glycans from individual samples, labeling each pool with a unique isobaric tag, combining the samples, and analyzing them in a single LC-MS/MS run. Quantification is achieved by measuring the intensity of reporter ions released during the MS/MS fragmentation.

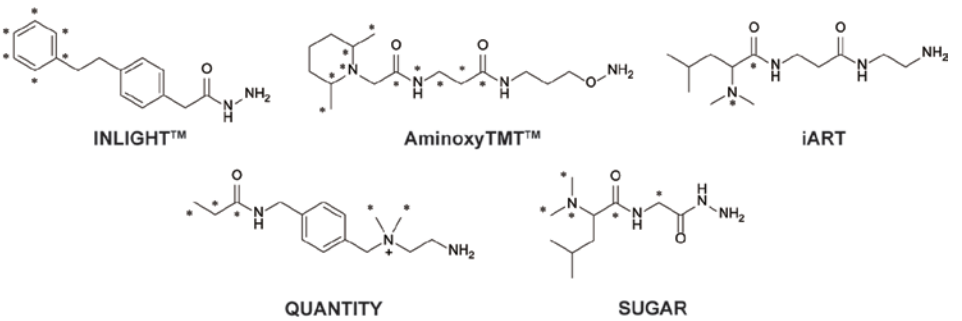


Figure 4.
Chemical structures of isobaric/isotopic tags developed for glycomics analysis. (A star denotes the position of isotopes used to create isobaric sets or light/heavy pairs.)

correction for isotopic overlap in high-mass glycans and a post-acquisition normalization method (Total Glycan Normalization Factor, TGNF) to correct for systematic variability more effectively than a single internal standard.

Commercialized by Cambridge Isotope Laboratories, Inc. and further developed, INLIGHT has been adapted for O-glycan and heparin oligomer analysis [23, 24], with evolving protocols and dedicated software [25–28]. Its applications have provided insights into areas such as xylem differentiation and ovarian cancer biomarker discovery [26, 29]. More recently, coupling INLIGHT with ion mobility spectrometry (IMS) has facilitated rapid glycan subclass identification based on molecular trend lines defined by m/z and collision cross-section (CCS) values [30].

3.2 AminoxyTMT™

A significant milestone was reached with the development of carbonyl-reactive tandem mass tags (glyco-TMTs) by Kuhn et al. in 2012, marking the first true isobaric reagents for glycans [31]. These tags adapted the proven TMT scaffold, testing both hydrazide and aminooxy functionalities as the reactive group. The aminooxy group was found to be superior, facilitating facile labeling of the glycan reducing end and demonstrating exceptional performance: labeling efficiencies greater than 95%, a coefficient of variation below 10%, and a dynamic range extending up to a 1:40 ratio. Such capabilities squarely address the critical need for robust quantification in glycomics research.

It is noteworthy that the initial study created not only duplex isobaric channels but also a light/heavy isotopic version, similar in principle to INLIGHT. This design allowed for a direct comparison of quantification methods within the same platform. The authors evaluated reporter ion quantification (the core isobaric tagging principle) against full MS quantification based on the area under the curve of the precursor ion. They concluded that under their experimental conditions, reporter ion quantification was less superior, primarily due to the inherent limitations of the tag chemistry discussed below.

Commercialized by Thermo Fisher Scientific as 6-plex AminoxyTMT, it remains the only commercially available isobaric tag specifically designed for glycan analysis. Its early application to N-glycan profiles from isogenic colorectal cancer cell lines SW480 and SW620 revealed significant differences between primary and metastatic tumors, underscoring its biomedical potential. Subsequent studies have leveraged it to identify differentiated glycan profiles in blood serum from patients with esophageal diseases [32], and dedicated software like SimGlycan has been developed to streamline data processing [33].

However, AminoxyTMT faces notable limitations that explain the performance gap observed in the original study. A primary challenge is the inefficiency of reporter ion generation. The TMT tag was originally optimized for peptide fragmentation, where the energy required to release the reporter ion matches that needed to break amide bonds. This energy is significantly higher than that required to fragment glycosidic bonds. Consequently, during MS/MS of glycans, energy is diverted into extensive carbohydrate fragmentation, leading to reduced recovery of the TMT reporter ions. Furthermore, the oxime bond formed by the aminooxy group is more labile than the stable secondary amine formed in reductive amination, increasing the probability of entire tag loss, which further suppresses reporter ion signal. The resulting low-reporter ion intensity and poor signal-to-noise ratio inevitably compromise quantification accuracy and dynamic range. Another practical issue is the interference

of the 126 Da reporter ion with a common HexNAc fragment (Δm = 0.07 Da), which cannot be resolved on medium-resolution instruments, effectively reducing the practical multiplexing capacity [31].

3.3 iART

Developed by Zhou et al. in 2013, isobaric aldehyde reactive tags (iARTs) represent an alternative strategy that leverages the well-established chemistry of reductive amination for glycan derivatization [34]. The hydrophobic nature of the iART structure enhances ionization efficiency in ESI-MS, yielding stronger signals even at lower glycan concentrations. The authors rigorously validated the quantitative accuracy of iARTs by demonstrating a strong consistency between the observed and expected ratios in labeled oligosaccharide standards, underscoring the reliability of the method for glycomic studies.

The practical utility of iARTs was demonstrated in disease research through the analysis of glycans from HIV gp120 glycoproteins, providing valuable insights into glycosylation patterns critical for understanding immune responses. Furthermore, the study highlighted the potential for expanding the iART technology to a six-plex set, which would enable simultaneous analysis of multiple samples and significantly improve throughput.

However, the study also acknowledged several limitations. A notable challenge is the requirement for higher collision energy to achieve complete fragmentation of larger glycans, which can compromise quantification accuracy. This is evidenced by higher coefficients of variation (CV) reported for glycans exceeding 2000 Da, indicating the reproducibility challenges for complex glycan structures. Additionally, as with any method relying on standard reductive amination, the labeling process under acidic conditions may lead to the loss of acid-labile sialic acids, potentially limiting the method's applicability for profiling sialylated glycans. Another limitation is the actual demonstrated throughput; while a six-plex expansion was proposed, the study only synthesized and practically validated a duplex set (channels 114 and 115). The variability in optimal fragmentation energy across different glycan sizes further complicates the method, necessitating detailed optimization for complex samples.

3.4 QUANTITY

As a successor to iART from the same research group, the QUANTITY method introduces a significant chemical innovation by incorporating a quaternary ammonium group into the isobaric tag [35]. This permanent positive charge addresses key challenges faced by its predecessor. It ensures enhanced ionization efficiency and detection sensitivity during MS analysis, making it particularly suitable for analyzing low-abundance glycans or limited sample quantities. Unlike methods that rely on metal ion adducts, QUANTITY avoids associated ion suppression issues. Most importantly, the permanent charge promotes efficient fragmentation during MS/MS, generating strong reporter ions for accurate quantification.

The study demonstrated the versatility of QUANTITY by achieving high labeling efficiency and applying it to the glycomic profiling of complex biological samples, including human serum and genetically engineered Chinese Hamster Ovary (CHO) cells. The method showed the excellent reproducibility and quantitative accuracy, with a strong linear correlation between theoretical and experimental ratios, underscoring its robustness.

One of the major strengths of QUANTITY is its seamless integration into existing workflows; it is compatible with the established protocols for 2-AA/2-AB labeling, allowing for easier adoption. However, the method has some limitations. Its multiplexing capacity is currently limited to four samples, which is lower than the commercially available 6-plex AminoxyTMT and may restrict its utility in very large-scale studies. Additionally, the reported protocol involves specialized steps, such as solid-phase glycoprotein immobilization and the use of specific reagents like *p*-toluidine for stabilizing sialic acids. These requirements could pose implementation challenges for laboratories lacking the necessary materials or technical expertise.

3.5 SUGAR

The SUGAR (Isobaric Multiplex Labeling Reagents for Carbonyl-Containing Compound) tags were developed as an economic and synthetically accessible alternative to TMT, building upon the compact dimethylated leucine (DiLeu) scaffold originally created for quantitative proteomics [36]. The structural simplicity of the DiLeu tag allows for straightforward synthetic extension, leading to a versatile family of reagents such as DiLeuPMP for O-glycan analysis [37], and DiLeuEN for glycopeptide and lipid analysis [38]. SUGAR tags employ a hydrazide functionality to target the glycan reducing end. Initially developed as a 4-plex set, the platform has been significantly expanded to a 12-plex format, representing the highest multiplexing capacity available for glycomics to date [39, 40].

The application of SUGAR tags has proven valuable in clinical and cell biology research. For instance, their use revealed that most N-glycans were down-regulated in B-cells from pediatric acute lymphoblastic leukemia patients after chemotherapy. A similar trend was observed in endometrial cancer cells treated with a quinone antimicrobial medication, highlighting the sensitivity of glycosylation as a biomarker for drug response. Furthermore, the platform's flexibility is demonstrated by its combination with a NeuCode strategy, which creates mass-defect versions of the tags. This allows for targeted analysis of low-abundance glycans by leveraging ultra-high-resolution mass spectrometers to differentiate mini-dalton mass differences [41, 42].

A critical performance metric for any isobaric tag is the efficiency of reporter ion generation. While the study demonstrated that SUGAR tags generate reporter ions more efficiently than AminoxyTMT, the intensity of these reporter ions was reported to be only around 10–20% of the most abundant oxonium ions (e.g., *m/z* 204 from HexNAc or *m/z* 366 from HexNAc-Man). This relative inefficiency can impact quantification sensitivity. Moreover, the low mass of the SUGAR reporter ions (below *m/z* 120) presents an analytical constraint: It necessitates setting a first mass cutoff during MS/MS method development. This setting prevents the detection of high-mass fragments that are crucial for glycan structural elucidation, potentially complicating data interpretation.

4. Limitations of current methods

The integration of isobaric labeling has undeniably accelerated high-throughput glycomics, creating a powerful synergy between novel probe chemistry, efficient derivatization, and advanced LC-MS/MS. As summarized in **Table 1**, the development of various tags has provided researchers with a toolkit for multiplexed analysis. Despite these significant advancements, a critical examination reveals several common limitations that represent both current challenges and opportunities for future innovation.

Tag name	Reactive chemistry	Multiplexing capacity	Key feature(s)	Primary limitation(s)
INLIGHT™	Hydrazide (reductive amination)	2-Plex (isotopic)	Pioneering isotopic tag; TGNF normalization; enhanced ionization	Not true isobaric multiplexing; limited to duplex analysis.
AminoxyTMT™	Aminooxy (oxime ligation)	6-Plex	First commercial isobaric tag; facile labelling and high labelling efficiency	Inefficient reporter ion generation; oxime bond lability; reporter ion interference (126 Da).
iART	Primary amine (reductive amination)	2-Plex (proposed 6-plex)	Hydrophobic tag improves ionization	High CV for large glycans; limited to duplex analysis (no follow-up to improve throughput)
QUANTITY	Primary amine (reductive amination)	4-Plex	Permanent charge enhances sensitivity and fragmentation	Limited throughput; protocol requires specialized steps (e.g., solid-phase immobilization).
SUGAR	Hydrazide (reductive amination)	4-Plex 12-Plex	Highest multiplexing capacity; based on synthetically accessible DiLeu scaffold	Low reporter ion intensity relative to oxonium ions; low first mass cutoff masks key structural ions.

Table 1.
Key characteristics of current isobaric and isotopic labeling platforms for glycomics, highlighting their respective advantages and limitations.

A fundamental constraint of current isobaric tags is their universal reliance on derivatizing the reducing end of glycans. Most tags, including iART, QUANTITY, and SUGAR, utilize reductive amination. This reaction requires acidic conditions and elevated temperature that are detrimental to acid-labile residues, notably sialic acids, leading to their loss and inaccurate profiling of sialylation [43]. Furthermore, the reaction produces salt byproducts that necessitate the extensive sample cleanup, complicating automation. While AminoxyTMT avoids reductive amination by forming an oxime bond, it introduces a different challenge. The reaction with the glycosylamine intermediate is fast, but if targeting the open-ring reducing end, it first requires a hydrolysis step to expose the reactive carbonyl group. This additional, often time-varying, hydrolysis can introduce analytical variance and lengthen sample preparation time.

As highlighted in the discussion of individual tags, a pervasive issue across all platforms is the inefficient generation of reporter ions. This stems from a fundamental mismatch in bond dissociation energies. The energy required to cleave the tag and release the reporter ion is typically optimized for peptide amide bonds. However, the glycosidic bonds of the glycan itself fragment at lower energies. Consequently, during MS/MS, collision energy is preferentially dissipated through extensive glycan fragmentation, producing dominant oxonium ions (often with further dehydration). The reporter ions are thus overshadowed, leading to lower signal-to-noise ratios. This directly translates to higher quantitative variance and a narrowed effective dynamic range compared to theoretical expectations.

Another gap could be filled in the further is the lack of dedicated, user-friendly bioinformatics platforms for data processing and interpretation. While the wet-lab workflow achieves high throughput, data analysis frequently becomes a manual, time-consuming bottleneck. Unlike in proteomics, where robust, integrated software suites seamlessly handle TMT or iTRAQ data from raw file to final quantitative ratios, the bioinformatics for isobaric glycan analysis often relies on a fragmented combination of proprietary instrument software, in-house scripts, and manual validation. This is compounded by the unique complexities of glycan data, including isobaric overlaps between different compositions, the presence of multiple charge states and adducts, and—crucially—the need to correctly associate MS2 spectra containing both quantitative reporter ions and structural carbohydrate fragments. The low intensity of reporter ions further exacerbates the challenge, requiring the sophisticated algorithms for signal extraction and noise filtering. However, integrated software solutions, such as PEAKS GlycanFinder, are continually evolving to address this technical gap [44].

5. Conclusion and future outlook

A significant opportunity for future development lies in bridging the gap between modern MS-based multiplexing and the established sensitivity of fluorescence detection. Classical high-throughput glycomics still heavily relies on fluorescent tags (e.g., 2-AB) coupled with UHPLC-FLD, which offers exceptional sensitivity and robustness against matrix effects. The current isobaric tags forego this capability.

The next generation of tags could be designed to incorporate a fluorophore alongside the isobaric mass tag. This would create a dual-detection platform where the same pooled sample could be analyzed first by UHPLC-FLD for highly sensitive, matrix-resistant quantification, and subsequently by LC-MS/MS for multiplexed, sample-specific quantification and structural confirmation. This orthogonal approach would leverage the strengths of both techniques, potentially boosting overall sensitivity, reliability, and data richness.

In conclusion, while isobaric labeling has propelled glycomics into a new era of throughput, addressing these limitations—through milder chemistries, optimized fragmentation schemes, and multi-modal tags—will be crucial for unlocking its full potential. The ongoing evolution of these tags will continue to deepen our understanding of the glycome in health and disease.

DOI: http://dx.doi.org/10.5772/intechopen.1013267

Author details

Miyang Li
Regeneron Pharmaceuticals, Inc., Tarrytown, NY, United States

*Address all correspondence to: miyangli88@gmail.com

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